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Dear Editorial Board,

Re: Submission of manuscript — "256×256 Original Hash: A Cryptographic Hash Function with Active Sacred Pair Inversion at Dimension 129"

I am writing to submit for your consideration the research article titled 256×256 Original Hash, the fifth work in an ongoing research program on pyramidal cryptographic hash functions. This algorithm represents the first design in the series where the Sacred Pair inversion — the K1→K2 boundary mechanism — is structurally active during normal operation, processing 127 of 255 pyramid anchors through the complementation space K2 in every permutation round.

The function maintains the original pyramidal anchor order (base to tip), a 256-byte internal state, and produces a 256-byte output through 24 permutation rounds. The Sacred Pair boundary is positioned at dimension 129: anchors at positions 0–127 remain in K1 (unchanged), while anchors at positions 128–254 cross into K2 and undergo bitwise complementation SP(x) = (~x & 0xFF). This active crossing of the K1/K2 boundary distinguishes the 256×256 Original Hash from all previous designs.

Eleven independent statistical tests on the complete 2048-bit output demonstrate: (1) avalanche effect of 49.804% with standard deviation of 1.196% over 150 samples; (2) Shannon entropy of 7.9975 bits/byte (99.97% efficiency); (3) Chi-square statistic of 268.77, well below the 95% critical value of 293.2; (4) Pearson correlation average of 0.02526 over 32,640 byte pairs at N=1,000 samples — statistically definitive independence; (5) Strict Avalanche Criterion deviation of 0.98% — the best SAC result in this research program; and (6) zero collisions in 400 attempts.

The SAC result of 0.98% is particularly noteworthy: only 161 of 16,384 output bit positions show deviation above 10% from ideal behavior. We attribute this exceptional result to the active K1/K2 transition at dimension 129, which acts as a structural epicenter of diffusion — introducing complemented values into 127 of 255 anchor positions in every round, creating a richer mixing landscape than any previous design in this series.

The article explicitly states the scope of this research: this is an experimental prototype for academic exploration. No claim of equivalence to established standards is made, and a formal cryptographic audit is recommended before any production use.

I sincerely thank the editorial committee for their time and consideration, and remain available for any clarification or additional information required.

Sincerely,

Marco Lazcano

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Project: 256×256 Original Hash — Research Program in Pyramidal Cryptography

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Manuscript enclosed: original_hash_256x256.docx